V1.4 — The Relational State Substrate

Reflex Engine, experiment V1.4 · OUTCOME D · 2026-09-01

The substrate does what it was built to do. A model cannot feed it.

On a sealed 600-case adversarial suite the substrate catches every seeded semantic error V1.3's representation could not see — 100% against 10%, both at zero false positives. On 240 sealed real-language items, driven by a model, it certifies **nothing at all**.

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1. Verdict against the frozen ladder

	condition	result
F	Tier A regressed, or C0 stops reproducing V1.3	**not triggered** — verified
D	any sealed suite above its wrong-fire gate	**TRIGGERED** — C1/C2/C3 at 100% wrong-fire before B22
C	substrate worse than C0 on semantic wrong-fire	not reached
B	H3 null at materially higher cost	not reached
A	H3 + (H4 or H5) + H9	**unreachable** — H9 unmeasured, H3 unassessable at zero served

OUTCOME D. Three sealed ablation arms served 17 answers on real language and every one was wrong (§4). The defect was found and fixed (B22); post-fix every arm serves zero and wrongs zero.

The ladder has a gap, and saying so is part of the result. After the fix no rung fits: D no longer fires (nothing is served, so nothing is wrong), C's literal condition fails (0% wrong-fire is not worse than C0's 1.44%), B describes a tie that did not happen, and A needs measurements that do not exist. The pre-registration never anticipated a system that is **safe and inert**. D is reported because it fired on a genuine sealed measurement during the run and because nothing above it is reachable — not because the corrected system is unsafe.

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2. The result that holds

Sealed adversarial suite, 600 cases, 10 semantic error classes, correct and corrupt state for every case, no model in the loop.

arm	detection	false positive
C0 — V1.3's flat plan IR	**10.0%**	0.0%

arm	detection	false positive
C2 — substrate + integrity	**100.0%**	0.0%

C0 catches one class. It misses incorrect grouping, entity mismatch, cross-entity contamination, wrong scope, temporal inversion, duplicate entities, missing state transitions and unresolved references — **because the flat IR discards the material those errors are about.**

Two deliberate generosities keep C0's 10% an upper bound rather than a convenient number: `agreement` was handed the same IR twice and always passes, and the projection preserves the plan DAG so groupings the flat IR *can* express are expressed.

Your `90 – (49 – 19)` case is **unrepresentable** in the substrate: the second event's argument names the derived node, so no edge lets 19 reach 49.

Reuse and equivalence, 6/6. All three structurally-similar / semantically-different pairs refused — including V1.3's Mrs Hilt / cupcakes failure — and all three lexically-different / semantically-equivalent pairs accepted. `gave away` vs `received` is refused at 0.615 text similarity, just under V1.3's 0.62 reuse threshold.

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3. The result that does not

240 sealed real-language items (S3, disjointness from all V1.3 usage proved by id intersection), produced by haiku through a bridge directory outside the repository (verified: no engine module reachable, no agent script referencing engine internals).

arm	served	correct	wrong	coverage
C1 substrate only	0	0	0	0.0%
C2 + integrity	0	0	0	0.0%
C3 + applicability	0	0	0	0.0%
C5 complete	**0**	**0**	**0**	**0.0%**

Where C5's coverage went:

```
agreement                           164   68.3%
integrity: wrong_target              42   17.5%
not_evaluable                        13    5.4%
integrity: missing_state_transition  10    4.2%
unknown_relation                      8    3.3%
integrity: entity_mismatch            1    0.4%
```

Against V1.3's flat IR on real language — 209 certified at 1.44% wrong

— this is a total collapse of coverage in exchange for the elimination of wrong answers. **The bottleneck moved rather than lifted:** from *checking* the reading to *producing* it.

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4. Seventeen wrong answers, and whose fault they were

Before the fix, the arms served 17 answers and every one was wrong. Not 17 mistakes — one mistake, 17 times:

```
"I see 349 frogs. How many legs do they have?"    served 349
"18 cubes long ... it doubles her length"         served 18
"Mrs. Hilt read 2 books per day ... in one week"  served 2
```

Each is a state with no resolved relation, whose only terminal is a stated given, executed and returned unchanged. The answer was the question's own input.

V1.3 caught this class. `check_nondegenerate` has guarded it since V1.3's own B4. I replaced the representation and did not re-derive what the old one was protecting against.

> ***A new representation inherits none of the old one's scar tissue.*** *Every guard the previous design accumulated has to be re-argued in the new terms, and the ones that get missed are exactly the ones nobody remembers needing.*

Fixed (B22); pre-fix arms preserved under `results/v14/arms/superseded_pre_B22/`.

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5. A correction to my own analysis

Agreement dominates C5's abstentions at 68.3%, so the first question was V1.3's D7 question: is the check rejecting equivalent readings over naming?

A decomposition said yes for 5 of 27 resolved items — operators matched while identity labels differed (`pigeons`/`pigeon`, `seeds_per_watermelon`/`quantity`). Labels are free text a producer chooses, so excluding them looked clearly right.

Excluding them recovered nothing. Failures went 19 → 20.

The decomposition was wrong. It compared *flattened* identity lists, so a sample that split a story into more relations appeared as a naming difference:

```
A = [seeds_per_watermelon, watermelons]   2 relations
B = [quantity, quantity, seeds, seeds]    4 relations
```

That is structural disagreement wearing the costume of a vocabulary disagreement. **Two samples of one model produce structurally different relational states for the same problem in roughly three quarters of resolved cases.**

The wrong reading was the comfortable one — it would have turned a hard finding about producer instability into a tidy comparison bug. Only the fix failing caught it.

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6. Hypotheses

	claim	result
H1	substrate improves semantic composition over the fixed representation	**CONFIRMED on the representation, FALSIFIED end-to-end**
H2	reduces wrong entity binding	**CONFIRMED** — 100% detection, 0% FP
H3	reduces structurally-valid-but-semantically-wrong computation	**CONFIRMED deterministically; UNASSESSABLE on the model path** (zero served)
H4	applicability conditions reduce semantic wrong-fire	**NOT EXERCISED** — 4 of 58 capabilities conditioned; C3 = C1 on every item
H5	integrity improves safety without unacceptable coverage loss	**HALF CONFIRMED** — safety yes; the coverage loss is total, so "unacceptable" is answered against it
H6	safe plan reuse becomes possible	**CONFIRMED on the confusion matrix**, 6/6
H7	improves Path Manager orchestration	**NOT RUN** (W6 deferred)
H8	improves real-language coverage	**FALSIFIED** — 69.7% (V1.3) → 0.0%
H9	preserves the end-to-end energy advantage	**FALSIFIED, and inverted** — 19.9x the GPU energy and 38.2x the CPU energy of a pure LLM, for zero correct answers (§8)
H10	stays tractable as complexity grows	**NOT RUN** (W9 deferred)

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7. Bugs (7)

- **B17** one-letter unit aliases matched English words — `"16 vases a day"` derived *amperes*.
- **B18** the physics lexicon has no everyday nouns, so derivation was UNKNOWN on nearly all real language and check 8 never fired — V1.3's D5 in a new place. Fixed with a separate lexicon harvested from **non-sealed** traffic.
- **B19** `m` is claimed by metres *and* molar; `"60 m of rope"` returned *concentration* because sort order picked a winner. **V1.3's B8 in a new layer.**
- **B20** a shell-mangled escape compiled a detector to match a backspace character — valid regex, no error, silently 0% detection.

- **B21** the inherited V1.3 stub filter discarded every valid V1.4 payload (it demanded `quantities/plan`, V1.4 sends `nodes/relations`).
- **B22** the substrate executor shipped without non-degeneracy — §4.
- **B23** a payload with the right keys and no content is still a stub. One agent wrote 59 of 60 as `{"nodes": [], "relations": []}` and called it "the foundational output file"; B21 had just widened the key check, so it passed. 121 purged. **Each fix to a stub filter teaches the next stub what to look like.**

The recurrence, now at five

```
V1.3 B16   consumption went silent on all-word problems
V1.3 B8    supersession starved the conflict check of its input
V1.4 B19   a colliding surface resolved by sort order
V1.4 B20   a mangled escape produced an inert regex
V1.4 B22   a guard that simply was not carried across a rewrite
```

A green check that cannot go red. Standing requirement adopted:

every detector needs a positive control — a case it MUST fire on —

running alongside the case it must not.

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8. Energy: the commercial thesis, measured and inverted

60 sealed items, local `llama3.1-custom`, one meter, one machine, both

arms on identical inputs. Every model call inside the boundary,

producer included.

	reflex (C5)	pure LLM	ratio
GPU J / query	**1,294.5**	64.9	**19.9x**
CPU J / query (modelled)	**6,850.6**	179.3	**38.2x**
correct answers / 60	**0**	**51**	—
J / correct (GPU)	**undefined**	76.4	—

Wall clock: 4,880 s for the reflex arm, 81 minutes for 60 questions.

The arc's commercial premise is that deterministic reflex execution

replaces expensive LLM reasoning and saves energy. Measured end to end,

this version does the opposite by more than an order of magnitude,

and returns nothing for it. J-per-correct is not a large number; it is

undefined, because the denominator is zero.

The mechanism is not mysterious. C5 makes two producer calls per query

at 900 max tokens each; the pure arm makes one call at 24. Roughly

seventy-five times the generated tokens, spent to build a state that

the gates then refuse.

Why V1.2's 21 mJ figure does not contradict this. That measured the

deterministic ENGINE with the model's work outside the window. It is

true and it answers a different question. Once the reading is inside

the boundary — and a deployment cannot put it outside — the model dominates the bill completely, and a design that calls the model MORE than the baseline cannot save energy no matter how efficient its arithmetic is.

The instrument was verified before these figures were believed: the first run reported 0.0 J on every channel (B24) and would have been the worst claim in the arc. A positive control now fails the run loudly rather than reporting a zero.

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9. What this version establishes

1. **The representation was never the hard part.** Moving from a flat plan IR to a relational substrate takes semantic-error detection from 10% to 100% at no false-positive cost. That is real and it is cheap.
2. **Producing the representation is the hard part.** Three quarters of dual samples disagree structurally; a fifth of surviving states answer a different question than the one asked. Neither is a checking problem.
3. **The gates work exactly as designed, and that is the problem.** Every abstention in §3 is defensible in isolation. Together they refuse everything.
4. **V1.3's flat IR is still the only configuration that has ever answered real language at scale** — 209 certified, 1.44% wrong. V1.4 is safer and useless; V1.3 is useful and imperfect. Nothing yet is both.
5. **The energy case does not survive putting the model inside the boundary.** Every prior energy claim in this arc measured the engine with the reading excluded. Measured whole, the architecture costs 20-38x a pure LLM and answers nothing. That is the finding a buyer would care about most, and it took until version sixteen to measure it honestly.

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10. What V1.4 does not claim

- **The energy figure is now measured and it is bad.** See §8. It is a single 60-item run on one machine with one local model, so the MAGNITUDE is provisional; the SIGN is not, and the sign is what the

commercial thesis rested on. H9 is unmeasured for the second version running, and it remains the commercial crux.

- **No cross-model comparison.** Only the haiku arm ran; sonnet, opus and llama-8B are unrun, so D-vi's architecture-and-reflex comparison is not delivered.
- **No applicability result.** 4 of 58 capabilities carry conditions and none fired on S3; C3 is identical to C1 on every item.
- **No Path Manager or tractability result** (W6, W9 deferred).

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11. The one-paragraph version

V1.4 replaced the representation and the bottleneck moved with it. The substrate holds every distinction the flat IR could not — 100% versus 10% semantic-error detection at zero false positives, the `90 – (49 – 19)` hazard made unrepresentable rather than merely caught, and V1.3's plan-reuse failure correctly refused — but on real language a model cannot produce states the substrate will accept, and the complete system certified nothing at all on 240 sealed items. The outcome is D, fired by seventeen wrong answers that turned out to be a guard I failed to carry across the rewrite. Worse for the project than any of that: measured end to end with the model inside the accounting boundary, the architecture spends 19.9x the GPU energy and 38.2x the CPU energy of simply asking the model, and gets nothing right while the model alone gets 51 of 60 — so the commercial premise of the whole arc is, on this evidence, inverted rather than merely unproven.